HUX / V1-PROOF / ELECTRICAL

Complete system wiring by location

Rear elevation of the R05 body. Robot LEFT (near wall) is at the left. Each board sits where it mounts. Ports are connector positions. Every harness conductor is drawn.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

LEFT LEG · NEAR WALL SIDE
fixture buttons, pack lead, left gearmotor

READING THE SHEET
Wire IDs → searchable conductor register.
Bundle cut lengths → EL-06 and the harness register.
Port labels: connector position / pin.
Dots connect. Gaps at crossings do not.
Dashed boards need part selection and bench checks.
BLACK wheel lead is switched OUT2, not ground.
XH = JST XH 2.50 mm. XT30 = 12 V power path.
1x6 0.1 in = the Pololu pigtail housing.

CTRL DECK · Z +170 · near half: IMU · far half: Pico 2

U1 / SparkFun LSM6DSO
3.3 V · main SPI pads · open the ADDR jumper for SPI · EL-02

JMAIN.5 / SDO — W002 — PH-C.1 / GP12
JMAIN.6 / CS — W003 — PH-C.2 / GP13
JMAIN.1 / GND — W004 — PH-C.3 / GND
JMAIN.4 / SCL — W005 — PH-C.4 / GP14
JMAIN.3 / SDA / SDI — W006 — PH-C.5 / GP15
JINT.2 / INT1 — PH-C.6 / GP16
JINT.1 / INT2
JMAIN.2 / 3V3 — W008 — PH-F.1 / 3V3(OUT)

SP1 splice

JMAIN: B6B-XH-A on the six main pads.
JINT: B2B-XH-A on INT2 / INT1.
INT2 stays empty.
Ports are in cable order; the position number gives the pad.

P1 / Pico 2
RP2350 · 3.3 V GPIO · eight XH headers in contiguous pin groups · GP25 LED onboard

PH-F.2 / 3V3_EN
PH-F.3 / GND
PH-F.4 / VSYS
PH-G.1 / GP0
PH-G.2 / GP1
PH-G.3 / GND

Headers: PH-A p4–p7, PH-A2 p8–p10, PH-B p11–p15, PH-C p16–p21, PH-D p24–p29, PH-E p31–p34, PH-F p36–p39, PH-G p1–p3.
Desolder the pre-soldered pins, then solder the XH headers to the board (2026-10-17).
NC: ADC_VREF p35, RUN p30, 3V3_EN p37, GP17 p22, GP19–GP21.
EL-02 shows the physical board and the IMU pins.

P1 / Pico 2

USB.1 / VBUS — W041 — CABLE.1 / VBUS
USB.2 / D+ — W042 — CABLE.2 / D+
USB.3 / D- — W043 — CABLE.3 / D-
USB.4 / GND — W044 — CABLE.4 / GND

RIGHT HARNESS TRUNK
down the front face of the body

PH-A.1 / GP2
PH-A.2 / GP3
PH-A.3 / GP4
PH-A.4 / GP5
PH-A2.1 / GND
PH-A2.2 / GP6 — W049
PH-A2.3 / GP7 — W050
PH-B.1 / GP8 — W054
PH-B.2 / GP9 — W055
PH-B.3 / GND — W053
PH-B.4 / GP10
PH-B.5 / GP11 — W051
PH-D.1 / GP18 — W056
PH-D.2 / GP19
PH-D.3 / GP20
PH-D.4 / GP21
PH-D.5 / GND — W058
PH-D.6 / GP22
PH-E.1 / GP26
PH-E.2 / GP27
PH-E.3 / AGND
PH-E.4 / GP28

RIGHT LEG · FAR WALL
USB cable, right gearmotor

JUSB / Laptop USB
CABLE.1 / VBUS
CABLE.2 / D+
CABLE.3 / D-
CABLE.4 / GND

Micro-USB cable to the front of the Pico 2.

W011
W010

BUS DECK · Z +120 · near half: power distribution · far half: spare cassette position

JPICO.1 / VSYS — JPICO.2 / GND

PDB / Power distribution

From the unswitched fused bus. D5 Schottky → VSYS.
Logic stays on through a kill. 5 V to JLEG.
5 V regulator · part TBD

J5V.1 / 5V
J5V.2 / GND
JIN.1 / BUS — JIN.2 / GND

SPARE CASSETTE POSITION
Was the servo bus and the encoder conversion. No board in V1-PROOF.
The encoder dividers sit on JLEG.

W046 W047 W048 W045
W009

PWR DECK · Z +60 · near half: kill chain · far half: driver board

JBUS.1 / BUS — JBUS.2 / GND
JCTL.1 / nSLEEP — JCTL.2 / 3V3

SWK / KILL
LEADS.1 / NO — W037 — JKILL.1 / 3V3K
LEADS.2 / COM — W038 — JKILL.2 / OFF node
red mushroom · momentary NO · fixture

SWA / ARM
LEADS.1 / NO — W039 — JARM.1 / 3V3K
LEADS.2 / COM — W040 — JARM.2 / ON node
green · momentary NO · fixture

BAT / 3S pack
LEADS.1 / + — RAIL.1 — W035 — XPACK.1 / +
LEADS.2 / - — RAIL.2 — XPACK.2 / -
on the tray · XT30 · TBD
inline holder · value TBD

The pack sits on the tray at the bottom of the body. It is drawn here so that its lead to the kill chain is clear.
Both buttons are on the bench fixture, not on the robot.

KC / Kill chain

Pololu 2813 + 74HC14 · EL-04
XPACK: fused pack in.
XRAIL: 2813 VOUT = the switched rail.
LD1117V33 gate logic from the bus.
100k / 22k rail divider → RAIL_ADC.
KILL → OFF. ARM → ON via 4.7k, clamped by HB_OK.
Not ordered or tested. K0–K6 open.

JHB.1 / HB
JHB.2 / RAIL_ADC
JHB.3 / AGND
XRAIL.1 / + — W014
XRAIL.2 / - — W015

DB / Driver board
DL, DR: Pololu 4035 / DRV8874 carriers on one 52 x 38 mm board · EL-03, EL-05

10 kΩ nFAULT pull-ups to 3V3. 1 kΩ CS series.
VREF divider ≈ 2.8 V (≈ 2.5 A). PMODE, IMODE to GND. VM NC.
nSLEEP from GP10 to both carriers.
EN low = brake. nSLEEP low = outputs off.
OUT1 / OUT2 go to JLEG, not to the motor directly.

XRAIL.1 / +
XRAIL.2 / -

JSIG-L.1 / EN
JSIG-L.2 / PH
JSIG-L.3 / nFAULT
JSIG-L.4 / CS out
JSIG-L.5 / GND
JSIG-R.1 / EN
JSIG-R.2 / PH
JSIG-R.3 / nFAULT
JSIG-R.4 / CS out
JSIG-R.5 / GND

JML.1 / OUT1 — JML.2 / OUT2 — JMR.1 / OUT1 — JMR.2 / OUT2

W060 W061 W062 W063

W016

UNDER THE DECKS · leg junction board under the PWR deck · 3S pack on the tray at −7 … −35

J5V.1 / 5V — J5V.2 / GND — JPL.1 / OUT1 — JPL.2 / OUT2 — JPR.1 / OUT1 — JPR.2 / OUT2 — JENC.1 — JENC.2 — JENC.3 — JENC.4 — JENC.5

ML / Pololu 4752 · LEFT WHEEL

ML
12 V · 30:1 · encoder

PIGTAIL.1 / RED — W022 — JML_IN.1 / RED
PIGTAIL.2 / BLACK — W023 — JML_IN.2 / BLACK
PIGTAIL.3 / GREEN — W024 — JML_IN.3 / GREEN
PIGTAIL.4 / BLUE — W025 — JML_IN.4 / BLUE
PIGTAIL.5 / YELLOW — W026 — JML_IN.5 / YELLOW
PIGTAIL.6 / WHITE — W027 — JML_IN.6 / WHITE

JLEG / Leg junction

Accepted 2026-10-17 · under the PWR deck · 1x6 pigtail headers · 2.2 kΩ / 3.3 kΩ dividers
Motor leads pass to JPL / JPR. Encoder A / B 5 V → 3.0 V → JENC.
Board layout, mount detail and the service loop for later leg motion remain open.

JMR_IN.1 / RED — W028 — PIGTAIL.1 / RED
JMR_IN.2 / BLACK — W029 — PIGTAIL.2 / BLACK
JMR_IN.3 / GREEN — W030 — PIGTAIL.3 / GREEN
JMR_IN.4 / BLUE — W031 — PIGTAIL.4 / BLUE
JMR_IN.5 / YELLOW — W032 — PIGTAIL.5 / YELLOW
JMR_IN.6 / WHITE — W033 — PIGTAIL.6 / WHITE

MR / Pololu 4752 · RIGHT WHEEL

MR
12 V · 30:1 · encoder

HUX / V1-PROOF / ELECTRICAL

Pico 2 + IMU connections

Component-side views, USB at the top of the Pico. Physical pins and GPIO numbers are distinct.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

PHYSICAL PIN MAP



p1 is upper left; p40 is upper right.

SEVEN-WIRE SENSOR HARNESS



SPI JUMPER SETUP / CHECK RECEIVED BOARD

Open the 0x6B / 0x6A address jumper so SDO can drive MISO.
Recommended: open both I2C pull-up traces. Keep SCX / SDIX ground jumpers intact.
Qwiic sockets carry I2C and are unused here. Wire the main SPI pads, not SCX / SDIX.
Power and signal logic are 3.3 V. Inspect board revision and check sensor identity before use.

HUX / V1-PROOF / ELECTRICAL

Wheel motor + encoder harness

One channel shown; repeat for the other wheel using the assignment table. Colors match the motor leads. The pigtail lands on the JLEG junction board.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

PICO / PROPOSED OUTPUTS

3.3 V command signals

Use the left / right GPIO table below.

POLOLU 4035 / TOP VIEW

VM: leave unused

Switched bus +

EN / PWM

PH / direction

nSLEEP (GP10)

VM
GND
EN / IN1
PH / IN2
PMODE
SLEEP
VREF

DRV 8874

VIN
GND
OUT1
OUT2
IMODE
FAULT
CS

RED

BLACK ≠ GND

POLOLU 4752 / MOTOR LEADS

12 V, 30:1 gearmotor

RED lead → OUT1
BLACK lead → OUT2
Both motor leads are switched.

PMODE + IMODE directly to GND

VREF divider from 3V3 ≈ 2.8 V: measure 2.5 A peak

FAULT / CURRENT INPUTS

Return to Pico on the driver board

nFAULT → GP11 / GP18; 10 kΩ pull-ups to 3V3
CS → GP26 / GP27 via 1 kΩ series
Latch faults; deliberate rearm required.

PICO / LEFT AND RIGHT CHANNELS

Proposed pin assignments

Function	Left	Right
Encoder A	GP2	GP4
Encoder B	GP3	GP5
EN / PH	GP6/7	GP8/9
FAULT	GP11	GP18
CS / ADC	GP26	GP27

JLEG JUNCTION BOARD / PROPOSED

5 V → 3.0 V resistor dividers

Four channels: 2.2 kΩ series, 3.3 kΩ to GND
Bench value from the electronics bench guide.
Measure the high level on GP2 before use.
No direct 5 V signals into Pico GPIO.

GND / GREEN
5 V / BLUE
A / YELLOW
B / WHITE

4752 / ENCODER LEADS

Six-wire vendor pigtail, 200 mm

GREEN → common GND
BLUE → regulated 5 V
YELLOW → channel A
WHITE → channel B
1x6 0.1 in housing → JLEG header. Verify by color.

EN low = brake. nSLEEP low = outputs off / coast. GP10 drives both nSLEEP pins.

2.5 A peak / 1.2 A RMS are Hux qualification targets, not tested settings.

HUX / V1-PROOF / ELECTRICAL

Power distribution + kill chain

Functional drawing of docs/kill-chain.md: fused pack, latched rail switch, heartbeat gate, logic branch. No part is ordered or tested.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

BATTERY / TBD
3S pack
Up to 12.6 V charged · XT30

F1 / INLINE
Main fuse
Value and wire gauge open

KC / POLOLU 2813
Latched rail switch
VIN → VOUT. ON / OFF pulse inputs.

RAIL

DB / ×2
DRV8874 VIN
Both carriers on the switched rail

BUS → logic

BUS → gate logic

OPEN ITEMS
Fuse value, rail wire gauge and the XT30 fit.
Return-energy clamp: the switch opens the supply path.
ON / OFF priority of the 2813 (stage K2).

KC / HEARTBEAT GATE
74HC14 + charge pump
GP22 500 Hz → 220 nF → BAT54S → 1 µF ‖ 100 kΩ
No edges for ≈ 80 ms → /HB_OK high → OFF
Own LD1117V33 3.3 V from the bus

PDB / UNSWITCHED LOGIC BRANCH
5 V regulator, part TBD
Fed before the switch: logic stays on through a kill
5 V to the encoders through JLEG

PDB / USB + EXTERNAL POWER
D5 Schottky → VSYS p39
External 5 V → diode → VSYS
Check the loaded voltage and USB backfeed

VSYS p39

P1 / CONTROLLER
Pico 2 power + sense
3V3(OUT) p36 → IMU + driver pull-ups
GP28 p34 ← rail divider 100 kΩ / 22 kΩ

KC / BUTTONS ON THE FIXTURE
KILL and ARM
KILL: 3.3 V → button → diode → OFF
ARM: 3.3 V → button → 4.7 kΩ → ON
ON clamped low while HB_OK is low
Second KILL press does nothing.
Rail returns only on ARM with the heartbeat.

SOFTWARE DISABLE, SEPARATE LAYER
GP10 p14 → both DRV8874 nSLEEP pins directly.
disarm and FAULT set nSLEEP low in the same cycle.
The rail switch is the second, independent layer.
A reset or reboot never turns the rail on.

P1 → KC / SENSE AND HEARTBEAT
Bundle B7, connector KC.JHB
GP22 p29 → JHB.1 HB
GP28 p34 ← JHB.2 RAIL_ADC
AGND p33 → JHB.3

HB / RAIL_ADC

COMMON GND / RETURNS

All GND symbols join this reference. Route the rail and the pack return away from the sensor and MCU wiring.

Rev C | 17 OCT 2026 | Vendor pinouts + proposed Hux circuits

See docs/wiring-atlas.md and the wire register for sources and open circuit decisions.

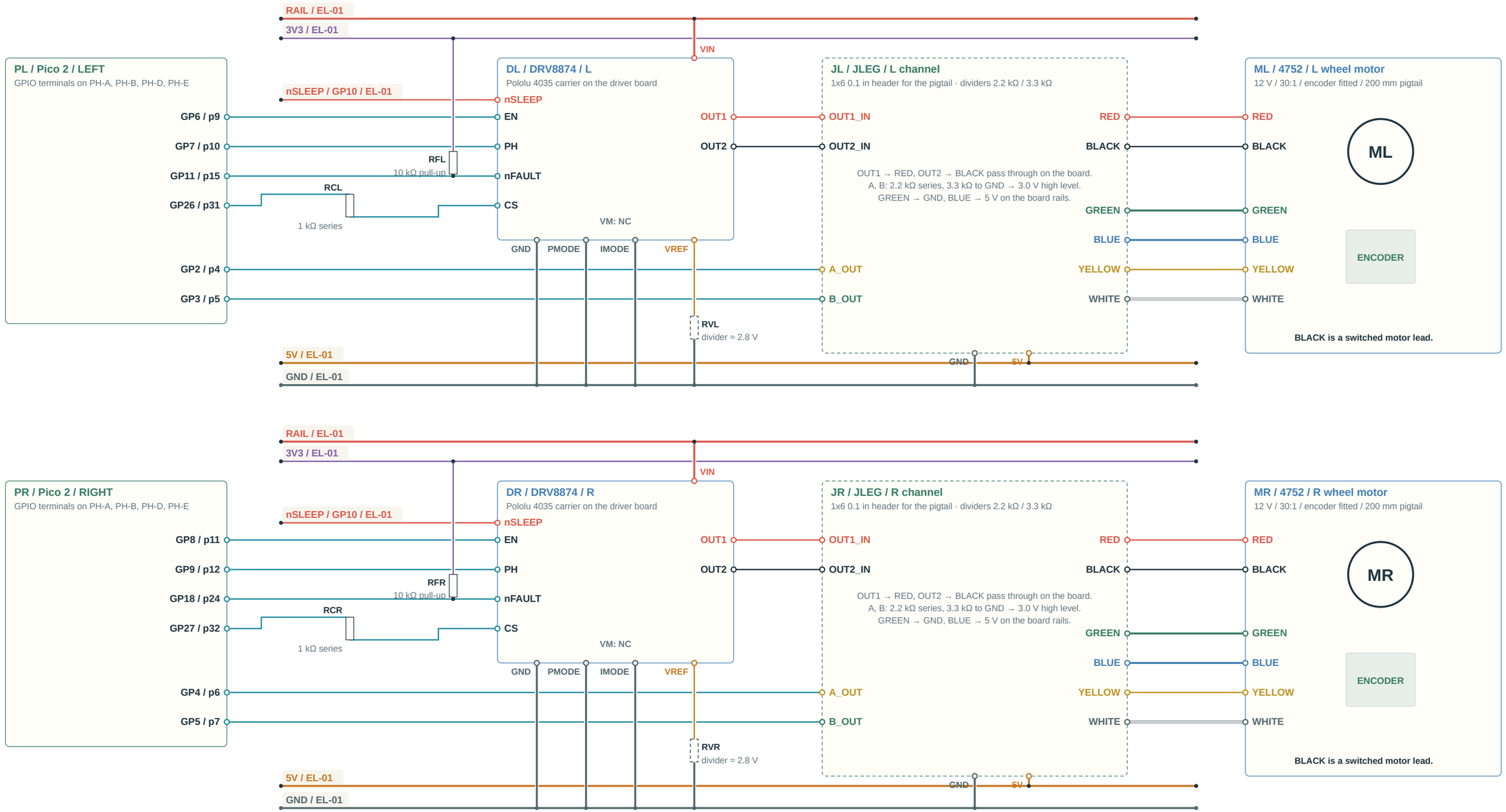
EL-04

HUX / V1-PROOF / ELECTRICAL

Both wheel motors — every wire

Two complete channels through the leg junction board. Rails and nSLEEP come from EL-01. Use terminal names here. EL-01 has the conductor register IDs.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN



Rev C | Source: 17 OCT 2026 | Print layout: 17 OCT 2026
17 x 11 in / landscape / 0.5 in margins / print at 100%. Sources and open checks: docs/wiring-atlas.md.

EL-05

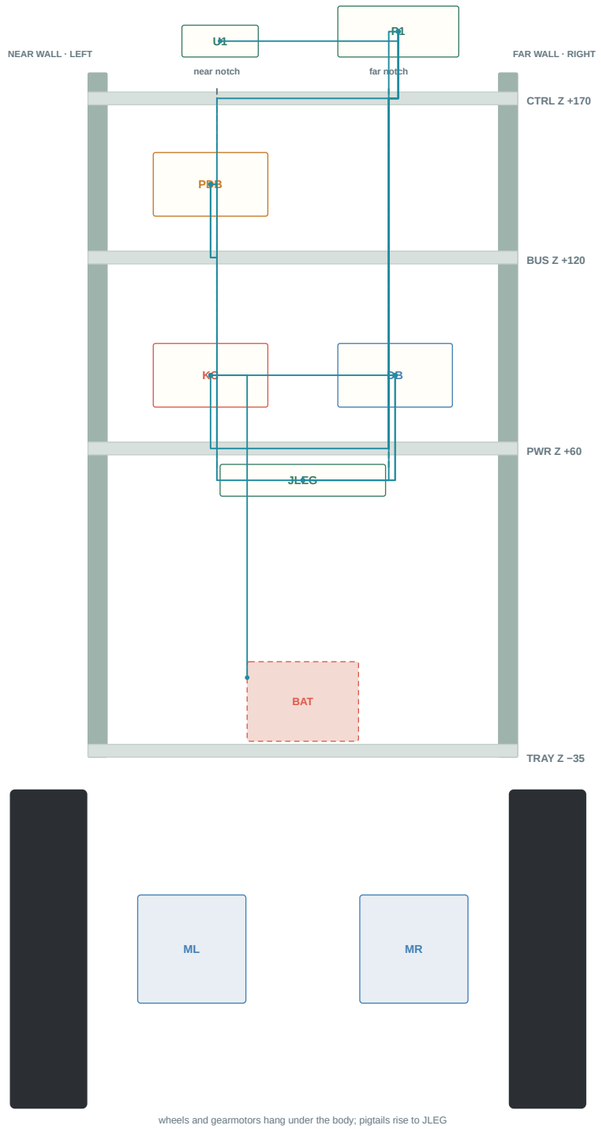
Harness bundles and run lengths

Route model on the R05 body, cut length for each bundle, connector hardware. Lengths are estimates: cut long, route on the frame, trim, record.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

ROUTE MODEL / FRONT ELEVATION

Boards at their R05 centres. Harness runs follow the front cable notches of each deck lane.



wheels and gearmotors hang under the body; pigtails rise to JLEG

BUNDLES / CUT LENGTHS

Bundle	Name	From → to	Wires	AWG	Model path	Cut length	Connectors
B1	**Sensor**	P1 → U1	7	24	59 mm	**95 mm**	P1.PH-C · SP1 · U1.JINT · U1.JMAIN
B2	**3V3 feed**	P1 → DB	2	22	275 mm	**335 mm**	DB.JCTL · P1.PH-F · SP1
B3	**Left driver signals**	P1 → DB	5	24	275 mm	**335 mm**	DB.JSIG-L · P1.PH-A2 · P1.PH-B · P1.PH-E
B4	**Right driver signals**	P1 → DB	5	24	275 mm	**335 mm**	DB.JSIG-R · P1.PH-B · P1.PH-D · P1.PH-E
B5	**Driver sleep**	P1 → DB	1	24	275 mm	**335 mm**	DB.JCTL · P1.PH-B
B6	**Encoder signals**	P1 → JLEG	5	24	287 mm	**350 mm**	JLEG.JENC · P1.PH-A · P1.PH-A2
B7	**Kill-chain sense**	P1 → KC	3	24	329 mm	**395 mm**	KC.JHB · P1.PH-D · P1.PH-E
B8	**Pico power**	PDB → P1	2	22	269 mm	**330 mm**	P1.PH-F · PDB.JPICO
B9	**Logic bus**	KC → PDB	2	22	226 mm	**280 mm**	KC.JBUS · PDB.JIN
B10	**Switched rail**	KC → DB	2	20	58 mm	**95 mm**	DB.XRAIL · KC.XRAIL
B11	**Encoder supply**	PDB → JLEG	2	22	238 mm	**295 mm**	JLEG.J5V · PDB.J5V
B12	**Left motor power**	DB → JLEG	2	22	62 mm	**100 mm**	DB.JML · JLEG.JPL
B13	**Right motor power**	DB → JLEG	2	22	62 mm	**100 mm**	DB.JMR · JLEG.JPR
B14	**Left motor pigtail (vendor)**	ML → JLEG	6	vendor	—	**200 mm fixed**	JLEG.JML_IN · ML.PIGTAIL
B15	**Right motor pigtail (vendor)**	MR → JLEG	6	vendor	—	**200 mm fixed**	JLEG.JMR_IN · MR.PIGTAIL
B16	**Pack**	BAT → KC	3	20	154 mm	**200 mm**	BAT.LEADS · F1.INLINE · KC.XPACK
B17	**KILL button (fixture)**	SWK → KC	2	24	—	**600 mm fixed**	KC.JKILL · SWK.LEADS
B18	**ARM button (fixture)**	SWA → KC	2	24	—	**600 mm fixed**	KC.JARM · SWA.LEADS
B19	**Laptop USB cable (external)**	JUSB → P1	4	cable	—	**—**	JUSB.CABLE · P1.USB

Length rule: Manhattan path: board -> front deck notch -> front face -> notch -> board. Cut = path x 1.10 + 2 x 15 mm, rounded up to 5 mm.
Totals: AWG 20 790 mm, AWG 22 2880 mm, AWG 24 9685 mm. Buy about 1.5 m of each.
Pigtail reach check: left motor exit to JLEG = 179 mm Manhattan, pigtail 200 mm. The left pigtail cannot reach the driver board (216 mm). JLEG accepted 2026-10-17.

CONNECTOR STANDARD

low_voltage: JST XH 2.50 mm: header B#B-XH-A, housing XHP-#, contact SXH-001T-P0.6 (22-28 AWG)
power: XT30 for the pack, the fuse and the switched rail
motor: Pololu 4752 pigtail keeps its 1x6 0.1 in housing; mates with a 1x6 male header on JLEG
pico_end: Decided 2026-10-17: the user desolders the pre-soldered pins of the Pico 2 and solders the XH headers to the board. The harness plugs into the Pico directly.
wire: 24 AWG signals, 22 AWG for 5 V / 3V3 / motor OUT1-OUT2, 20 AWG for the pack and the rail

HARDWARE COUNT (V1-PROOF harness)

Housings: XHP-2 × 15 XHP-3 × 3 XHP-4 × 3 XHP-5 × 4 XHP-6 × 3 B2B-XH-A × 13
Headers: B3B-XH-A × 3 B4B-XH-A × 3 B5B-XH-A × 4 B6B-XH-A × 3
Contacts SXH-001T-P0.6: 81 used, buy 106. XT30 pairs: 2. 1x6 0.1 in male headers: 2. Crimp tool: IWISS SN-01BM or Engineer PA-09.

Connector loading

Every harness connector with its positions. XH = JST XH 2.50 mm. Position 1 is the header pin nearest the triangle mark on the housing; mark pin 1 on each board.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

P1.PH-A
XH · 4 positions · p4-p7 · Encoder inputs

1	GP2	L_ENC_A	H021 → JLEG.JENC.1 (B6)
2	GP3	L_ENC_B	H022 → JLEG.JENC.2 (B6)
3	GP4	R_ENC_A	H023 → JLEG.JENC.3 (B6)
4	GP5	R_ENC_B	H024 → JLEG.JENC.4 (B6)

P1.PH-A2
XH · 3 positions · p8-p10 · Encoder ground, left EN / PH

1	GND	GND	H025 → JLEG.JENC.5 (B6)
2	GP6	L_EN	H010 → DB.JSIG-L.1 (B3)
3	GP7	L_PH	H011 → DB.JSIG-L.2 (B3)

P1.PH-B
XH · 5 positions · p11-p15 · Right EN / PH, nSLEEP, left FAULT

1	GP8	R_EN	H015 → DB.JSIG-R.1 (B4)
2	GP9	R_PH	H016 → DB.JSIG-R.2 (B4)
3	GND	GND	H014 → DB.JSIG-L.5 (B3)
4	GP10	nSLEEP	H020 → DB.JCTL.1 (B5)
5	GP11	L_FAULT	H012 → DB.JSIG-L.3 (B3)

P1.PH-C
XH · 6 positions · p16-p21 · SPI1 sensor

1	GP12	SPI_SDO	H001 → U1.JMAIN.5 (B1)
2	GP13	SPI_CS	H002 → U1.JMAIN.6 (B1)
3	GND	GND	H003 → U1.JMAIN.1 (B1)
4	GP14	SPI_SCL	H004 → U1.JMAIN.4 (B1)
5	GP15	SPI_SDI	H005 → U1.JMAIN.3 (B1)
6	GP16	IMU_INT1	H006 → U1.JINT.2 (B1)

P1.PH-D
XH · 6 positions · p24-p29 · Right FAULT, heartbeat

1	GP18	R_FAULT	H017 → DB.JSIG-R.3 (B4)
2	GP19	empty	
3	GP20	empty	
4	GP21	empty	
5	GND	GND	H019 → DB.JSIG-R.5 (B4)
6	GP22	HB	H026 → KC.JHB.1 (B7)

P1.PH-E
XH · 4 positions · p31-p34 · Current sense, rail sense

1	GP26	L_CS	H013 → DB.JSIG-L.4 (B3)
2	GP27	R_CS	H018 → DB.JSIG-R.4 (B4)
3	AGND	AGND	H028 → KC.JHB.3 (B7)
4	GP28	RAIL_ADC	H027 → KC.JHB.2 (B7)

P1.PH-F
XH · 4 positions · p36-p39 · Logic power

1	3V3(OUT)	3V3	H008 → SP1 (B2)
2	3V3_EN	empty	
3	GND	GND	H030 → PDB.JPICO.2 (B8)
4	VSYS	VSYS	H029 → PDB.JPICO.1 (B8)

P1.PH-G
XH · 3 positions · p1-p3 · Optional UART0 link (no cable in V1-PROOF)

1	GP0	empty	
2	GP1	empty	
3	GND	empty	

P1.USB
Micro-USB · 4 positions · onboard · Onboard Micro-USB

1	VBUS	USB_VBUS	H060 → JUSB.CABLE.1 (B19)
2	D+	USB_D+	H061 → JUSB.CABLE.2 (B19)
3	D-	USB_D-	H062 → JUSB.CABLE.3 (B19)
4	GND	GND	H063 → JUSB.CABLE.4 (B19)

U1.JMAIN
XH · 6 positions · main pads · LSM6DSO main SPI pads

1	GND	GND	H003 → P1.PH-C.3 (B1)
2	3V3	3V3	H007 → SP1 (B1)
3	SDA / SDI	SPI_SDI	H005 → P1.PH-C.5 (B1)
4	SCL	SPI_SCL	H004 → P1.PH-C.4 (B1)
5	SDO	SPI_SDO	H001 → P1.PH-C.1 (B1)
6	CS	SPI_CS	H002 → P1.PH-C.2 (B1)

U1.JINT
XH · 2 positions · INT2, INT1 pads · LSM6DSO interrupt pads

1	INT2	empty	
2	INT1	IMU_INT1	H006 → P1.PH-C.6 (B1)

DB.JSIG-L
XH · 5 positions · protoboard · Left driver signals

1	EN	L_EN	H010 → P1.PH-A2.2 (B3)
2	PH	L_PH	H011 → P1.PH-A2.3 (B3)
3	nFAULT	L_FAULT	H012 → P1.PH-B.5 (B3)
4	CS out	L_CS	H013 → P1.PH-E.1 (B3)
5	GND	GND	H014 → P1.PH-B.3 (B3)

DB.JSIG-R
XH · 5 positions · protoboard · Right driver signals

1	EN	R_EN	H015 → P1.PH-B.1 (B4)
2	PH	R_PH	H016 → P1.PH-B.2 (B4)
3	nFAULT	R_FAULT	H017 → P1.PH-D.1 (B4)
4	CS out	R_CS	H018 → P1.PH-E.2 (B4)
5	GND	GND	H019 → P1.PH-D.5 (B4)

DB.JCTL
XH · 2 positions · protoboard · Shared nSLEEP and pull-up supply

1	nSLEEP	nSLEEP	H020 → P1.PH-B.4 (B5)
2	3V3	3V3	H009 → SP1 (B2)

DB.JML
XH · 2 positions · protoboard · Left motor OUT1 / OUT2

1	OUT1	L_MOTOR_OUT1	H037 → JLEG.JPL.1 (B12)
2	OUT2	L_MOTOR_OUT2	H038 → JLEG.JPL.2 (B12)

DB.JMR
XH · 2 positions · protoboard · Right motor OUT1 / OUT2

1	OUT1	R_MOTOR_OUT1	H039 → JLEG.JPR.1 (B13)
2	OUT2	R_MOTOR_OUT2	H040 → JLEG.JPR.2 (B13)

DB.XRAIL
XT30 · 2 positions · pigtail · Switched rail in

1	+	RAIL	H033 → KC.XRAIL.1 (B10)
2	-	GND	H034 → KC.XRAIL.2 (B10)

JLEG.JML_IN
P6 · 6 positions · 1x6 male header · Left Pololu pigtail

1	RED	L_MOTOR_OUT1	H041 → ML.PIGTAIL.1 (B14)
2	BLACK	L_MOTOR_OUT2	H042 → ML.PIGTAIL.2 (B14)
3	GREEN	GND	H043 → ML.PIGTAIL.3 (B14)
4	BLUE	5V	H044 → ML.PIGTAIL.4 (B14)
5	YELLOW	L_ENC_A5	H045 → ML.PIGTAIL.5 (B14)
6	WHITE		H046 → ML.PIGTAIL.6 (B14)

JLEG.JMR_IN
P6 · 6 positions · 1x6 male header · Right Pololu pigtail

1	RED	R_MOTOR_OUT1	H047 → MR.PIGTAIL.1 (B15)
2	BLACK	R_MOTOR_OUT2	H048 → MR.PIGTAIL.2 (B15)
3	GREEN	GND	H049 → MR.PIGTAIL.3 (B15)
4	BLUE	5V	H050 → MR.PIGTAIL.4 (B15)
5	YELLOW	R_ENC_A5	H051 → MR.PIGTAIL.5 (B15)
6	WHITE		H052 → MR.PIGTAIL.6 (B15)

JLEG.JPL
XH · 2 positions · protoboard · Left motor power to driver

1	OUT1	L_MOTOR_OUT1	H037 → DB.JML.1 (B12)
2	OUT2	L_MOTOR_OUT2	H038 → DB.JML.2 (B12)

JLEG.JPR
XH · 2 positions · protoboard · Right motor power to driver

1	OUT1	R_MOTOR_OUT1	H039 → DB.JMR.1 (B13)
2	OUT2	R_MOTOR_OUT2	H040 → DB.JMR.2 (B13)

JLEG.JENC
XH · 5 positions · protoboard · Encoder outputs, 3.3 V level

1	L_A	L_ENC_A	H021 → P1.PH-A.1 (B6)
2	L_B	L_ENC_B	H022 → P1.PH-A.2 (B6)
3	R_A	R_ENC_A	H023 → P1.PH-A.3 (B6)
4	R_B	R_ENC_B	H024 → P1.PH-A.4 (B6)
5	GND	GND	H025 → P1.PH-A2.1 (B6)

JLEG.J5V
XH · 2 positions · protoboard · Encoder supply

1	5V	5V	H035 → PDB.J5V.1 (B11)
2	GND	GND	H036 → PDB.J5V.2 (B11)

KC.XPACK
XT30 · 2 positions · pigtail · Fused pack in

1	+	PACK_FUSED	H054 → F1.INLINE.2 (B16)
2	-	GND	H055 → BAT.LEADS.2 (B16)

KC.XRAIL
XT30 · 2 positions · pigtail · Switched rail out (Pololu 2813 VOUT)

1	+	RAIL	H033 → DB.XRAIL.1 (B10)
2	-	GND	H034 → DB.XRAIL.2 (B10)

KC.JBUS
XH · 2 positions · protoboard · Unswitched fused bus to the 5 V regulator

1	BUS	BUS	H031 → PDB.JIN.1 (B9)
2	GND	GND	H032 → PDB.JIN.2 (B9)

KC.JHB
XH · 3 positions · protoboard · Heartbeat in, rail sense out

1	HB	HB	H026 → P1.PH-D.6 (B7)
2	RAIL_ADC	RAIL_ADC	H027 → P1.PH-E.4 (B7)
3	AGND	AGND	H028 → P1.PH-E.3 (B7)

KC.JKILL
XH · 2 positions · protoboard · KILL button

1	3V3K	KILL_A	H056 → SWK.LEADS.1 (B17)
2	OFF node	KILL_B	H057 → SWK.LEADS.2 (B17)

KC.JARM
XH · 2 positions · protoboard · ARM button

1	3V3K	ARM_A	H058 → SWA.LEADS.1 (B18)
2	ON node	ARM_B	H059 → SWA.LEADS.2 (B18)

PDB.JIN
XH · 2 positions · protoboard · Fused bus in

1	BUS	BUS	H031 → KC.JBUS.1 (B9)
2	GND	GND	H032 → KC.JBUS.2 (B9)

PDB.JPICO
XH · 2 positions · protoboard · 5 V through D5 to VSYS

1	VSYS	VSYS	H029 → P1.PH-F.4 (B8)
2	GND	GND	H030 → P1.PH-F.3 (B8)

PDB.J5V
XH · 2 positions · protoboard · 5 V to the encoder junction

1	5V	5V	H035 → JLEG.J5V.1 (B11)
2	GND	GND	H036 → JLEG.J5V.2 (B11)

SWK.LEADS
XH · 2 positions · button · KILL button leads

1	NO	KILL_A	H056 → KC.JKILL.1 (B17)
2	COM	KILL_B	H057 → KC.JKILL.2 (B17)

SWA.LEADS
XH · 2 positions · button · ARM button leads

1	NO	ARM_A	H058 → KC.JARM.1 (B18)
2	COM	ARM_B	H059 → KC.JARM.2 (B18)

ML.PIGTAIL
P6 · 6 positions · vendor housing · Pololu 4752 leads, 200 mm

1	RED	L_MOTOR_OUT1	H041 → JLEG.JML_IN.1 (B14)
2	BLACK	L_MOTOR_OUT2	H042 → JLEG.JML_IN.2 (B14)
3	GREEN	GND	H043 → JLEG.JML_IN.3 (B14)
4	BLUE	5V	H044 → JLEG.JML_IN.4 (B14)
5	YELLOW	L_ENC_A5	H045 → JLEG.JML_IN.5 (B14)
6	WHITE		H046 → JLEG.JML_IN.6 (B14)

MR.PIGTAIL
P6 · 6 positions · vendor housing · Pololu 4752 leads, 200 mm

1	RED	R_MOTOR_OUT1	H047 → JLEG.JMR_IN.1 (B15)
2	BLACK	R_MOTOR_OUT2	H048 → JLEG.JMR_IN.2 (B15)
3	GREEN	GND	H049 → JLEG.JMR_IN.3 (B15)
4	BLUE	5V	H050 → JLEG.JMR_IN.4 (B15)
5	YELLOW	R_ENC_A5	H051 → JLEG.JMR_IN.5 (B15)
6	WHITE		H052 → JLEG.JMR_IN.6 (B15)

Pack leads, the inline fuse and the laptop USB cable are not listed; they are not harness connectors. The three empty Pico positions in PH-D keep GP19–GP21 free for a later revision.